



The impact of schoolyard interventions on children's and adolescents' social, emotional, and cognitive well-being: A scoping review

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ABSTRACT

Background: Growing mental well-being concerns in children and adolescents, combined with the underexplored potential of schoolyards to positively impact well-being, highlight the need for further knowledge in this area. The objective of this scoping review was to provide an overview of existing research on the impact of schoolyard modifications on social, emotional, and cognitive well-being of children and adolescents.

Methods: Following the PRISMA-ScR guidelines, the inclusion criteria comprised children and adolescents in primary or secondary education and interventions involving modifications to the physical schoolyard environment (i.e. play structures, markings, and greening). Quantitative before/after evaluations of interventions were eligible. Five databases (Embase, Web of Science, SPORTDiscus, PsycINFO, and ERIC) were searched in July 2024. Quality of the included studies was assessed using an adapted version of the Effective Public Health Practice Project (EPHPP) Quality Assessment Tool.

Results: Fourteen papers, evaluating 13 studies, were included, of which five were rated 'strong' in quality, three were rated 'moderate,' and five were rated 'weak'. Thirty-one outcomes using 18 different measurement tools were identified, which highlights the complexity of the well-being construct. Most studies yielded non-significant or conflicting findings. However, findings cautiously indicated that a greater variety of play structures, spending time in green areas, and play structures combined with educational components tended to improve social, emotional, and cognitive well-being. These features should be prioritised when modifying schoolyards.

Conclusion: The evidence base remains limited and fragmented, which highlights the need for further research based on robust study designs and sample sizes sufficient to detect differences.

1. Introduction

Global estimates indicate that 5–14 % of 2–19-year-olds experience mental well-being challenges. (Health Resources & Services Administration [HRSA], 2020; Lorth et al., 2023; World Health Organization [WHO], 2025), which makes the development of interventions designed to improve child and adolescent mental well-being a public health priority (Patel et al., 2007). In this study mental health is conceptualised through the three domains; social, emotional, and cognitive well-being based on the definition proposed by Galderisi et al. (2015). Improving mental well-being is complex and requires multi-level interventions. While clinical and therapeutic approaches are essential, there is growing

recognition of the value of preventative strategies, as early interventions during childhood and adolescence can reduce the onset of mental illness later in life (Arango et al., 2018). Improving schoolyards through modification, i.e., deliberate changes to playground design and equipment, provides an unique opportunity to promote mental well-being, as nearly 90 % of all children and adolescents attend school, most of which have a surrounding schoolyard (United Nations Children's Fund [UNICEF], 2022). Ensuring that schoolyards are safe, accessible, fun, and motivating, enables outdoor play and other activities that supports mental well-being through risk-taking, resilience, and social interaction (Dodd et al., 2023; Gray, 2011).

The health benefits of meeting the recommended daily 60-minutes of

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moderate to vigorous physical activity in 4–17-year-olds are well documented (Guthold et al., 2018; Poitras et al., 2016), with evidence showing a reduction in mental well-being challenges, including anxiety and depression (Ferreira et al., 2020; Janssen & LeBlanc, 2010; Rodriguez-Ayllon et al., 2019). In the schoolyard, peer interaction and unstructured play provide opportunities to solve problems, regulate emotions, and manage behaviour, all of which support social well-being (Burdette & Whitaker, 2005; Gray, 2011; Rademacher & Koglin, 2019). Risky play, such as climbing or rough-and-tumble activities, further enhances mental well-being by fostering coping strategies, and building resilience (Dodd & Lester, 2021; Sandseter, 2007). Together, these skills act as protective factors against anxiety and depression and contribute to improved overall mental well-being (Meagher et al., 2009).

Although research has examined the impact of schoolyards on various aspects of mental well-being (Bikomeye et al., 2021; Broekhuizen et al., 2014; Schipperijn et al., 2024), none has specifically focused on modifications while adopting a broad definition of well-being. Given the significant, yet underexplored, potential of schoolyards to positively influence well-being, the objective of this scoping review was to provide an overview of scientific research investigating the impact of schoolyard modifications on children's and adolescents' social, emotional, and cognitive well-being. Identifying available research and mapping how mental well-being has been measured, as well as which constructs have been used, can inform researchers in conducting practice-relevant studies that may lead to improved mental well-being in children and adolescents.

2. Method

The study was initially planned as a systematic review, with a protocol registered in the International Prospective Register of Systematic Reviews (CRD42024562600) in June 2024. However, due to the diversity of outcomes, a formal synthesis was not feasible. Given a limited and heterogeneous evidence base, we determined that a scoping review (Munn et al., 2018) would be more appropriate for mapping existing well-being constructs and measurement approaches. This scoping review was conducted following guidelines from The Preferred Reporting Items for Systematic Review and Meta-Analysis extension for Scoping Reviews (PRISMA-ScR, Page et al., 2021; see Appendix A for checklist).

2.1. Eligibility criteria

The eligibility criteria followed the Population, Intervention, Comparison, Outcomes, and Study design (PICOS) framework (Higgins et al., 2019).

2.1.1. Population

Children and adolescents aged 4 to 17 years in primary or secondary education. Other populations were excluded unless measured alongside this target group.

2.1.2. Intervention

Eligible studies had to include physical environment modifications in schoolyards. A recent scoping review identified play structures, markings, and greening as relevant modifications (Schipperijn et al., 2024). However, schoolyard modifications are often implemented alongside structural or educational interventions. To reflect everyday school life, we included additional interventions if they were implemented alongside modifications to the physical environment.

2.1.3. Comparison

Control conditions included any intervention other than modifications to the physical environment. Uncontrolled studies were also eligible, if they had a pre-post design.

2.1.4. Outcome

Outcomes pertaining to mental well-being at the child and adolescent level, operationalised as social, emotional, or cognitive well-being were included. Social well-being involves interpersonal aspects such as relationships, interactions, and social behaviour. Emotional well-being relates to intrapersonal factors like feelings and thoughts, and cognitive well-being refers to mental functions such as learning, memory, and problem-solving (Galderisi et al., 2015).

2.1.5. Study design

Eligible designs included peer-reviewed randomised controlled trials, cross-over studies, controlled before-and-after studies, uncontrolled before-and-after studies, and repeated cross-sectional studies. Studies with only qualitative measures were excluded.

2.2. Search strategy

The literature search was conducted in July 2024 using five databases: Embase (Ovid), Web of Science (Clarivate), SPORTDiscus (EBSCOhost), PsycINFO (Ovid), and ERIC (EBSCOhost). Scopus was initially considered but excluded based on librarian recommendation, due to the low likelihood of identifying studies not already present in the other databases. A comprehensive search was reviewed by the research team who all have expertise in the field, refined with help of a research librarian, and tested by the first author. The final search strategy was adapted for each database. The search was not limited by language or publication year. The search strategies for all databases can be found in Appendix B.

2.3. Selection process

Eligible studies were uploaded to EndNote 20.1, where duplicates were removed. All records were then imported into ASReview LAB version 1.6.2 for title/abstract screening. This machine learning tool has proven efficient in prioritising records based on their relevance to the search (Campos et al., 2023; Schoot et al., 2021), and has been used in similar studies (Nigg et al., 2024; Zhang et al., 2023). ASReview continuously prioritises records during manual screening within the software, reducing the need for manual screening. A detailed overview of this process is presented in Appendix C.

The screening process was conducted by two independent groups of researchers (Group 1: IL, CSP; Group 2: AJ, OT, PM) for comparative analysis. Each group member screened in three-hour blocks, then passed the task to the next member, continuing until the stopping criterion was met. The stopping criterion was defined as 6 % of consecutive records being identified as irrelevant. The 6 % threshold was selected based on simulation studies of similar research topics (Boetje & van de Schoot, 2024; Campos et al., 2024; Ferdinands et al., 2023).

After title/abstract screening in ASReview, full-text articles were manually screened for inclusion using Covidence (www.covidence.org/). Each article was independently assessed by two researchers (IL, CSP, AJ, or PM), and when two reviewers disagreed, another reviewer resolved the disagreement (IL or CSP).

2.4. Data extraction and data items

The Covidence data extraction tool was customised to include necessary data items as outlined in Covidence's practical guide for data extraction (Covidence, 2024). In addition to basic information, we extracted the characteristics of study, the study design and methods, participant details, intervention type, outcomes measured, and results reported. Two reviewers (CSP, IL) independently extracted the data. Disagreements were resolved through discussion.

2.5. Quality assessment

The quality of all included studies was assessed using the relevant components of the Effective Public Health Practice Project (EPHPP) Quality Assessment Tool (McMaster Evidence Review & Synthesis Team [MERST], 2024; Thomas et al., 2004). The EPHPP tool is widely used in public health research to assess the methodological quality of individual studies. Since our review focused on school-based interventions, we deemed that assessing 'blinding', 'intervention integrity' and 'analyses' using the EPHPP questions was not meaningful or applicable for this review. The interventions we were interested in were conducted at school level, were very visible to both children and researchers, and recording individual levels of exposure to the intervention was not feasible. Two reviewers (AJ, PM) independently assessed the quality at the study level, resolving disagreements through discussion. See Appendix D for the version of the EPHPP we used in this review.

2.6. Synthesis of extracted results

A meta-analysis was initially considered but could not be conducted due to the large variation in outcome measures and diverging outcome reporting. It was not meaningful to report effect sizes either due to lack and inconsistency of reporting in the included articles. For these reasons, synthesising all extracted results in one analysis was not meaningful either (McKenzie & Brennan, 2024). Instead, we summarised the results reported in the included studies, grouped into three main outcome categories: Social, emotional, and cognitive well-being. However, this structure was not used until the discussion section to avoid repetition, as most studies assessed outcomes related to more than one well-being category. For each study, we also assessed type of intervention implemented. Finally, we decided to prioritise the results of the higher quality studies by presenting their outcomes in greater detail.

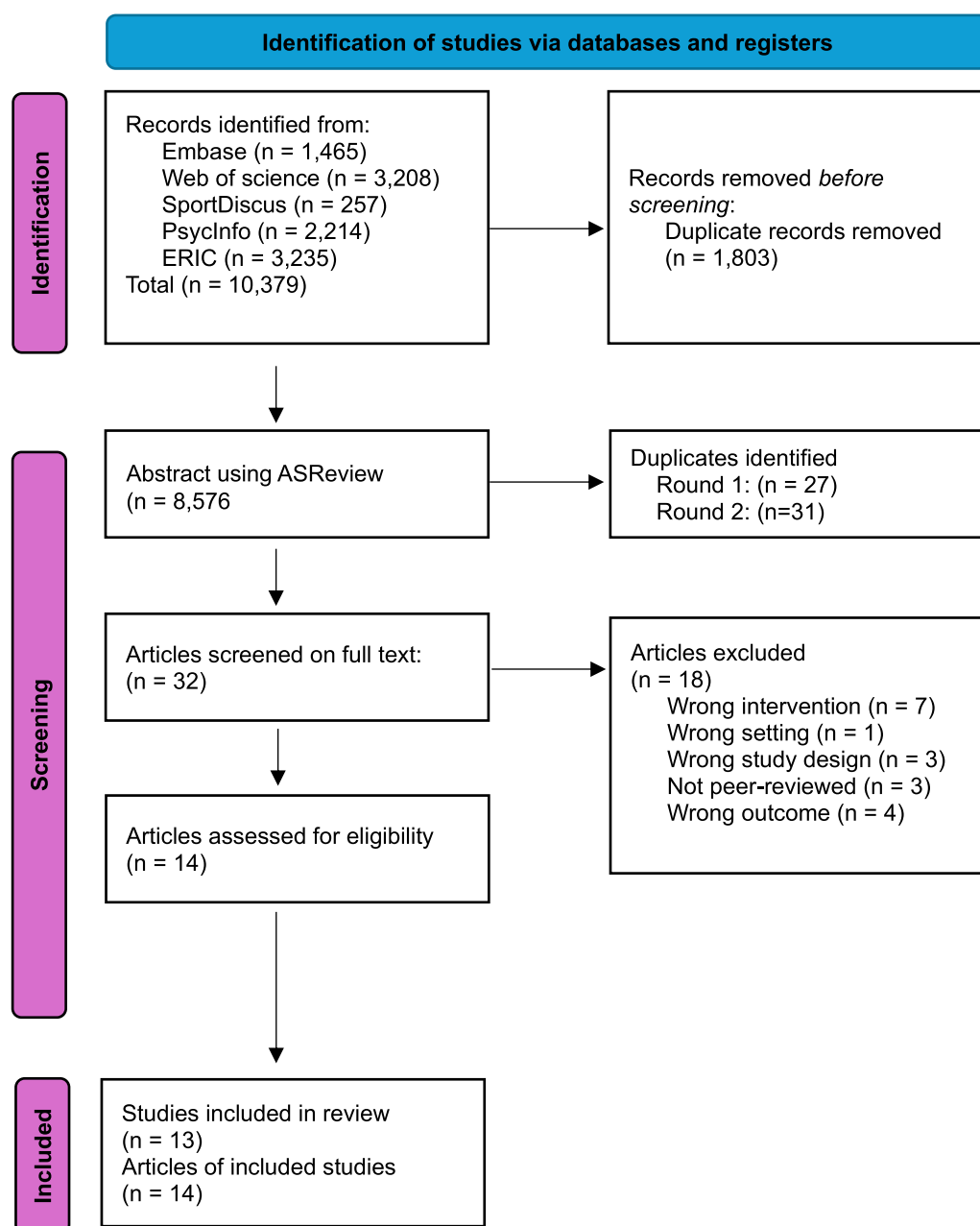


Fig. 1. PRISMA flowchart.

Table 1
Characteristics of included studies.

Author	Year	Country	Study design	Number of schools (intervention + controls)	Number of participants	Population age and characteristics	Intervention of primary interest	Other Elements of the Intervention
Bohnert AM, Nicholson LM, Mertz L, Bates CR, Gerstein DE	2022	USA	Uncontrolled before-and-after study	Primary school, (2 intervention schools)	980 pupils, 178/164 caregivers, school staff, community members pre/post intervention	prekindergarten (3.6 %), kindergarten through fourth grade (46.2 %), fifth through eighth grade (36.6 %), high school (0.8 %), and adults (12.9 %)	Play structures + Greening	
Crust L, McKenna J, Spence J, Thomas C, Evans D, Bishop D	2014	England	Controlled before-and-after study	Primary school, (12 intervention schools + 4 control schools)	218 pupils	10–11 yrs.	Markings	Education/facilitation
Farmer VL, Williams SM, Mann JI, Schofield G, McPhee JC, Taylor RW	2017	New Zealand	Randomized controlled trial	Primary school, 8 intervention schools + 8 control schools)	840 pupils, 635 parents, 90 teachers	6–7 yrs. + 8–9 yrs,	Play structures + Greening	Loose equipment + Fewer rules
Griffin AA, Caldarella P, Sabey CV, Heath MA	2017	USA	Repeated cross-sectional	Primary school, (1 intervention school with 2 playgrounds)	388 pupils	5–11 yrs	Play structures, Education/facilitation	
Kelz C, Evans GW, Röderer K,	2015	Austria	Controlled before-and-after study	Secondary school, (1 intervention school + 2 control schools)	133 pupils	13–15 yrs.	Play structures + Greening	
Khan M, McGeown S, Bell S,	2020	Bangladesh	Controlled before-and-after study	Primary school, (1 intervention school with both treatment and control group + 1 control school)	123 pupils	8–11 yrs.	Play structures	Education/facilitation
Lateef S, Zahir R, Sherdil L, McCleary C, Shafin T,	2014	India	Repeated cross-sectional	Primary school, (1 intervention school)	140 pupils (girls)	5–11 yrs.	Play structures + Markings	
Mayfield CA, Child S, Weaver RG, Zarrett N, Beets MW, Moore JB,	2017	USA	Repeated cross-sectional	Primary school, (2 intervention schools + 2 control schools)	not reported	8–11 yrs.	Markings	Loose equipment + Education/facilitation
Raney MA, Hendry CF, Yee SA,	2019	USA	Controlled before-and-after study	Primary school, (1 intervention school + 1 control school)	931 pupils	5–11 yrs	Greening	
Raney MA, Bowers AL, Rissberger AL,	2021	USA	Same as Raney (2019)	Same as Raney (2019)	1031 pupils	Same as Raney (2019)	Same as Raney (2019)	
Redondo-Tebar A, Ruiz-Hermosa A, Martínez-Vizcaino V, Bermejo-Cantarero A, Cavero-Redondo I, Martín-Espinosa NM, Sanchez-Lopez M,	2023	Spain	Randomized controlled trial	Primary school + Preschool, (11 intervention schools + 10 control schools)	1168 pupils	4–6 yrs.	Play structures	Loose equipment + Education/facilitation
Sánchez-López M, Cavero-Redondo I, Álvarez-Bueno C, Ruiz-Hermosa A, Pozuelo-Carrascosa DP, Díez-Fernández A, Gutierrez-Díaz del Campo D, Pardo-Guijarro MJ, Martínez-Vizcaíno V,	2019	Spain	Randomized controlled trial	Primary school, (4 intervention schools + 5 control schools)	240 pupils	5–7 yrs.	Play structures	Loose equipment + Education/facilitation
van Dijk-Wesselius, JE, Maas J, Hovinga D, van Vugt M, van den Berg AE,	2018	The Netherlands	Controlled before-and-after study	Primary school, (5 intervention schools + 4 control schools)	706/682/643 pupils pre/post/second post intervention	7–11 yrs.	Greening	
Yuill N, Strieth S, Roake C, Aspden R, Todd B	2007	England	Uncontrolled before-and-after study	Special school, (1 intervention school)	8 pupils	5–7 yrs., autistic spectrum disorder	Play structures	New rules

3. Results

3.1. Results of the literature search

Details on the number of records included and excluded throughout the screening process are visualised in Fig. 1. The initial search across the five databases yielded 10,379 records. After removing 1803 duplicates in EndNote, title and abstracts of 8576 records were screened using ASReview. Using a 6 % stopping criterion, we stopped screening after identifying 515 consecutive irrelevant records. For Group 1, this occurred after screening 11.2 % of the dataset, while for Group 2, the percentage screened was 8.8 %. Group 1 identified 31 records as relevant, while Group 2 identified 20. Merging the datasets resulted in 32 unique records. Of the 32 articles screened at full-text level, 14 articles, representing 13 studies, were eligible for inclusion.

3.2. Study characteristics

Characteristics of included studies are described in detail in Table 1. The articles were published between 2007 and 2024.

3.2.1. Place and population

Four studies were from the USA (Bohnert et al., 2022; Griffin et al., 2017; Mayfield et al., 2017; Raney et al., 2021; Raney et al., 2019), two each from Spain (Redondo-Tebar et al., 2023; Sánchez-López et al., 2019), and England (Crust et al., 2014; Yuill et al., 2007), and one each from New Zealand (Farmer et al., 2017), The Netherlands (van Dijk-Wesselius et al., 2018), Austria (Kelz et al., 2015), Bangladesh (Khan et al., 2020), and India (Lateef et al., 2024). Eleven studies were conducted in primary schools, with one also including preschool pupils (Redondo-Tebar et al., 2023). One study was conducted in a secondary school (Kelz et al., 2015), and one in a school for children with autistic spectrum disorders (Yuill et al., 2007). Most participants were aged five to 13 years.

3.2.2. Sample size

A total of 87 schools were represented, either as intervention or control schools. The number of children participating at each school varied. One study (Yuill et al., 2007) included eight children with autism spectrum disorders from one school, the other studies ranged in sample size from 123 to 1168 children, attending between 1 and 21 schools. For further details, see Table 1.

3.2.3. Interventions

Nine interventions modified play structures in schoolyards, such as installing climbing frames, adding soccer goals, or placing buddy benches (Bohnert et al., 2022; Farmer et al., 2017; Griffin et al., 2017; Kelz et al., 2015; Khan et al., 2020; Lateef et al., 2024; Redondo-Tebar et al., 2023; Sánchez-López et al., 2019; Yuill et al., 2007). Five interventions included greening initiatives, such as increasing access to vegetation by adding shrubs, potted plants, trees, and grass (Bohnert et al., 2022; Farmer et al., 2017; Kelz et al., 2015; Raney et al., 2021; Raney et al., 2019; van Dijk-Wesselius et al., 2018). Three interventions involved adding playground markings (Crust et al., 2014; Lateef et al., 2024; Mayfield et al., 2017). Four studies evaluated a combination of these three types of modifications (Bohnert et al., 2022; Farmer et al., 2017; Kelz et al., 2015; Lateef et al., 2024), while eight studies also included additional components, such as educational programs, facilitation, provision of unfixed equipment, or fewer schoolyard rules (Crust et al., 2014; Farmer et al., 2017; Griffin et al., 2017; Khan et al., 2020; Mayfield et al., 2017; Redondo-Tebar et al., 2023; Sánchez-López et al., 2019; Yuill et al., 2007). For further details see Table 1.

3.2.4. Study designs and measurements

The study designs included randomised controlled trials ($n = 3$; Farmer et al., 2017; Redondo-Tebar et al., 2023; Sánchez-López et al.,

2019), controlled before-and-after studies ($n = 5$; Crust et al., 2014; Kelz et al., 2015; Khan et al., 2020; Raney et al., 2021; Raney et al., 2019; van Dijk-Wesselius et al., 2018), uncontrolled before-and-after studies ($n = 2$; Bohnert et al., 2022; Yuill et al., 2007), or repeated cross-sectional studies ($n = 3$; Griffin et al., 2017; Lateef et al., 2024; Mayfield et al., 2017). Four types of measurements were used: Surveys ($n = 8$; Bohnert et al., 2022; Crust et al., 2014; Farmer et al., 2017; Kelz et al., 2015; Khan et al., 2020; Lateef et al., 2024; Redondo-Tebar et al., 2023; van Dijk-Wesselius et al., 2018), systematic observations ($n = 5$; Bohnert et al., 2022; Crust et al., 2014; Griffin et al., 2017; Mayfield et al., 2017; Raney et al., 2021; Raney et al., 2019), performance-based tasks ($n = 4$; Kelz et al., 2015; Sánchez-López et al., 2019; van Dijk-Wesselius et al., 2018; Khan et al., 2020), or camera recordings ($n = 1$; Yuill et al., 2007). Five studies used more than one type of measurement (Bohnert et al., 2022; Crust et al., 2014; Kelz et al., 2015; Khan et al., 2020; van Dijk-Wesselius et al., 2018). In total, 18 different measurement instruments were used, of which 12 were reported to be validated. The only measurement instrument that was used in more than one study was the System for Observing Children's Activity and Relationships during Play (SOCARP), which was used in four studies (Bohnert et al., 2022; Crust et al., 2014; Mayfield et al., 2017; Raney et al., 2021; Raney et al., 2019). See Table 2 for details.

3.3. Quality of studies

Of the 13 eligible studies, five were rated 'strong' (Kelz et al., 2015; Raney et al., 2021; Raney et al., 2019; Redondo-Tebar et al., 2023; Sánchez-López et al., 2019; van Dijk-Wesselius et al., 2018), three were rated 'moderate' (Farmer et al., 2017; Khan et al., 2020; Yuill et al., 2007), and five were rated 'weak' (Bohnert et al., 2022; Crust et al., 2014; Griffin et al., 2017; Lateef et al., 2024; Mayfield et al., 2017). The five studies rated 'strong' shared a strong study design (RCT or controlled before and after study), used a validated measurement instrument, and included adequate control for confounding in their design/analyses. The quality scores for each study can be found in Table 3.

3.4. Outcomes reported

In total, 13 studies looked at 31 different outcomes. Eleven studies (Bohnert et al., 2022; Crust et al., 2014; Farmer et al., 2017; Griffin et al., 2017; Khan et al., 2020; Lateef et al., 2024; Mayfield et al., 2017; Raney et al., 2021; Raney et al., 2019; Redondo-Tebar et al., 2023; van Dijk-Wesselius et al., 2018; Yuill et al., 2007) examined 16 different outcomes related to social well-being (e.g., social cohesion, prosocial/antisocial behaviour, and peer problems), using surveys, structured observations, video observations, and performance-based tasks. Five studies (Crust et al., 2014; Kelz et al., 2015; Lateef et al., 2024; Redondo-Tebar et al., 2023; van Dijk-Wesselius et al., 2018) used surveys to assess seven outcomes related to emotional well-being (e.g., self-esteem, self-perception, and emotional functioning). Finally, four studies (Kelz et al., 2015; Khan et al., 2020; Sánchez-López et al., 2019; van Dijk-Wesselius et al., 2018) used performance-based tasks to measure eight different outcomes related to cognitive well-being (e.g., executive functioning, academic attainment, and information processing speed). See Table 2 for complete list.

3.4.1. Studies of strong quality

The results of the five 'strong' studies were reported in six articles (Kelz et al., 2015; Raney et al., 2021; Raney et al., 2019; Redondo-Tebar et al., 2023; Sánchez-López et al., 2019; van Dijk-Wesselius et al., 2018). Two studies (Redondo-Tebar et al., 2023; Sánchez-López et al., 2019) evaluated the effects of different parts of the multicomponent RCT called MOVI-KIDS conducted in Spain on nine outcomes related to social, emotional, and cognitive well-being. The intervention involved changes of play structures in the schoolyard combined with adding loose equipment, facilitation of activities after school hours, as well as

Table 2

Overview of outcomes and types of measurement per study.

Outcome category	Outcome	Type of measurement	Measurement instrument	Validated tool	Bohnert et al. (2022)	Crust et al. (2014)	Farmer et al. (2017)	Griffin et al. (2017)	Kelz et al. (2015)	Khan et al. (2020)	Lateef et al. (2024)	Mayfield et al. (2017)	Raney et al. (2019, 2021)	Redondo-Tebar et al. (2023)	Sánchez-López et al. (2019)	Van Dijk-Wesselius et al. (2018)	Yuill et al. (2007)
Social wellbeing	Social cohesion	Survey	study specific	Can't tell	1												
	Social interaction	Systematic observation	SOCARP	Yes	1												
	Prosocial behaviour	Systematic observation	SOCARP	Yes		1						1					
	Antisocial behaviour	Systematic observation	SOCARP	Yes								1	1				
	Prosocial interactions	Systematic observation	SOCARP	Yes									1				
	Peer relations	Survey	The Peer Relations Assessment Questionnaires, revised	No			1										
	Solitary behaviour	Systematic observation	Systematic observation	Can't tell				1									
	Perceived collaboration	Survey	study specific	Yes						1							
	Social well-being items	Survey	study specific	No							1						
	Friends	Survey	health-related quality of life (HRQoL) questionnaire	Yes										1			
	Prosocial orientation	Performance-based tasks	The Social Orientation Choice Card)	Yes												1	
	Prosocial behaviour	Survey	subscales from the Strength and Difficulties Questionnaire	Yes												1	
	Peer problems	Survey	subscales from the Strength and Difficulties Questionnaire	Yes												1	
	Social support	Survey	subscale called 'social support in friendships'	Can't tell												1	
	Play behaviour	Video recordings	Systematic rating of behaviors	No													1
	Social initiations	Video recordings	Systematic rating of behaviors	No													1
Emotional wellbeing	Physical self-perception	Survey	Children and Youth Physical Self-Perception Profile – CY-PSPP	Yes		1											
	Intra-psycho balance	Survey	The Basler Well-Being Questionnaire	Can't tell					1								
	Emotional well-being items	Survey	study specific	No							1						

(continued on next page)

Table 2 (continued)

Outcome category	Outcome	Type of measurement	Measurement instrument	Validated tool	Bohnert et al. (2022)	Crust et al. (2014)	Farmer et al. (2017)	Griffin et al. (2017)	Kelz et al. (2015)	Khan et al. (2020)	Lateef et al. (2024)	Mayfield et al. (2017)	Raney et al. (2019, 2021)	Redondo-Tebar et al. (2023)	Sánchez-López et al. (2019)	Van Dijk-Wesselius et al. (2018)	Yuill et al. (2007)
Cognitive wellbeing	Quality of life	Survey	health-related quality of life (HRQoL) questionnaire	Yes										1			
	Emotional well-being	Survey	health-related quality of life (HRQoL) questionnaire	Yes										1			
	Self-esteem	Survey	health-related quality of life (HRQoL) questionnaire	Yes										1			
	Emotional functioning	Survey	subscale from the Paediatric Quality of life scale	Yes												1	
	Executive functioning	Performance-based tasks	The Attention Network Test	Yes					1								
	Academic attainment	Performance-based tasks	Exam results in math + science	Yes						1							
	Logical reasoning	Performance-based tasks	General and Differential Aptitudes for schoolchildren aged 6-8 yrs	Yes											1		
	Verbal factor	Performance-based tasks	General and Differential Aptitudes for schoolchildren aged 6-8 yrs	Yes											1		
	Numerical factor	Performance-based tasks	General and Differential Aptitudes for schoolchildren aged 6-8 yrs	Yes											1		
	Spatial factor	Performance-based tasks	General and Differential Aptitudes for schoolchildren aged 6-8 yrs	Yes											1		
	General intelligence	Performance-based tasks	General and Differential Aptitudes for schoolchildren aged 6-8 yrs	Yes											1		
	Attentional functioning	Performance-based tasks	Digit Letter Substitution Test + Sky Search task	Yes												1	

parental encouragement. Redondo-Tebar et al. showed significant improvements in boys' emotional well-being, but not for girls, and non-significant effects for social well-being for both boys and girls (Redondo-Tebar et al., 2023). Sánchez-López et al. found significant improvements in cognitive well-being (Sánchez-López et al., 2019).

The controlled before-and-after study by Raney et al. (Raney et al., 2021; Raney et al., 2019) evaluated the effects of a playground greening project on social well-being at one school in Los Angeles (USA) using systematic observations. The study reported significant improvements in prosocial interactions. Antisocial behaviour did not change significantly from before to immediately after greening but decreased significantly at 4-month follow-up at the experimental school (Raney et al., 2019). At 16 months post-intervention, a persistent significant positive change was found for the intervention group in antisocial behaviour and prosocial interactions (Raney et al., 2021).

Using a controlled before-and-after design, the Dutch study by van Dijk-Wesselius et al. (2018) looked at six different outcomes related to both social, emotional, and cognitive well-being, and mostly found non-significant results after greening schoolyards at five schools. However, intervention schools improved significantly in self-reported social support and peer problems decreased. No difference in prosocial behaviour or emotional functioning was reported. No differences in cognitive well-being were observed at the first follow-up, however, at the second follow-up, two years post-intervention, the intervention group significantly outperformed the control group (2018).

The controlled before-and-after study in Austria by Kelz et al. (2015) assessed emotional and cognitive well-being among 13–15-year-old pupils. The results showed that pupils at the intervention school scored significantly higher on intrapsychic balance after the installation of the new schoolyard (Kelz et al., 2015). Both the intervention school and two control schools demonstrated significant improvements in executive functioning, with no differences observed between them after implementing a new schoolyard with more green elements and new play structures at the intervention school (Kelz et al., 2015).

3.4.2. Studies of moderate quality

Results from the three studies rated to be of 'moderate quality' (Farmer et al., 2017; Khan et al., 2020; Yuill et al., 2007) were also mixed. In an RCT in New Zealand, Farmer et al. (2017) reported mixed findings regarding social well-being following a schoolyard modification that included play structures, greening, loose equipment, and reduced schoolyard rules. On one hand, peer play improved, and parents noted that intervention children were more likely to have happy relationships with other pupils at one year. On the other hand, teachers in intervention schools observed a higher frequency of physical bullying after two years, but there were no differences in children reporting bullying incidents to teachers at either one or two years (Farmer et al., 2017).

In an uncontrolled study in England, Yuill et al. (2007) investigated social well-being in eight children with autism spectrum disorders

attending a school for children with additional support needs. The children transitioned to a schoolyard specifically designed to meet their needs and behaviour changes were studied using video recordings. Video recordings revealed a significant decrease in solitary play and a significant increase in group play, both of which are linked to social well-being (Yuill et al., 2007).

The controlled before-and-after study by Khan et al. (2020) in Bangladesh investigated the impact of a schoolyard modification featuring a garden and amphitheatre on cognitive well-being. The study included a control school and a control group within the intervention school, where pupils used the new facilities but did not attend outdoor lectures. Only pupils who received math and science lessons in the modified spaces showed significant improvements in academic attainment compared to both control conditions (Khan et al., 2020).

3.4.3. Studies of week quality

The remaining five studies (Bohnert et al., 2022; Crust et al., 2014; Griffin et al., 2017; Lateef et al., 2024; Mayfield et al., 2017) were rated as 'weak', primarily because they had a repeated-cross-sectional design, or an uncontrolled before and after design. Griffin et al. (2017) conducted systematic observations to assess solitary play, an indicator of social well-being. Their findings showed a significant reduction in the number of children engaging in solitary play following installation of buddy benches and instructions on how to use them (Griffin et al., 2017). Bohnert et al. (2022) studied the effect of a combination of adding new play structures and greening in the USA on social well-being and observed a lower proportion of negative social interactions and neutral social interactions, and a higher proportion of positive interactions. Social cohesion did not change (Bohnert et al., 2022).

Three studies by Crust and et al. (2014); Mayfield et al. (2017), and Lateef et al. (2024) evaluated schoolyard interventions where new playground markings were added. Crust et al. (2014) investigated the effects of markings and trained supervisors on social and emotional well-being, observing a 6.7 % increase in prosocial behaviours. Self-perception did not change significantly (Crust et al., 2014). Lateef et al. (2024) reported different outcomes with conflicting findings in emotional and social well-being following a schoolyard modification including both markings and play structures (Lateef et al., 2024). Mayfield et al. (2017) also found different outcomes with conflicting findings in social well-being when examining an intervention involving markings, loose equipment, and education. They observed primarily non-significant result on antisocial and prosocial behaviour between groups and over time.

4. Discussion

4.1. Social well-being

The effect of new play structures was the most frequently studied

Table 3

Study quality scores assessed using the relevant components of the Effective Public Health Practice Project (EPHPP) Quality Assessment Tool. For more information see Appendix D.

Author	Selection bias	Study design	Confounders	Data collection methods	Withdrawals and drop-outs	Overall rating
Bohnert et al. (2022)	3 = Weak	2 = Moderate	1 = Strong	3 = Weak	3 = Weak	3 = Weak
Crust et al. (2014)	3 = Weak	1 = Strong	3 = Weak	2 = Moderate	3 = Weak	3 = Weak
Farmer et al. (2017)	3 = Weak	1 = Strong	1 = Strong	3 = Weak	1 = Strong	2 = Moderate
Griffin et al. (2017)	3 = Weak	3 = Weak	3 = Weak	3 = Weak	3 = Weak	3 = Weak
Kelz et al. (2015)	3 = Weak	1 = Strong	1 = Strong	2 = Moderate	2 = Moderate	1 = Strong
Khan et al. (2020)	3 = Weak	1 = Strong	1 = Strong	1 = Strong	3 = Weak	2 = Moderate
Lateef et al. (2024)	3 = Weak	3 = Weak	3 = Weak	3 = Weak	3 = Weak	3 = Weak
Mayfield et al. (2017)	3 = Weak	3 = Weak	3 = Weak	1 = Strong	3 = Weak	3 = Weak
Raney et al. (2019, 2021)	3 = Weak	1 = Strong	2 = Moderate	1 = Strong	1 = Strong	1 = Strong
Redondo-Tebar et al. (2023)	2 = Moderate	1 = Strong	1 = Strong	1 = Strong	1 = Strong	1 = Strong
Sánchez-López et al. (2019)	2 = Moderate	1 = Strong	1 = Strong	1 = Strong	1 = Strong	1 = Strong
Van Dijk-Wesselius et al. (2018)	3 = Weak	1 = Strong	1 = Strong	2 = Moderate	1 = Strong	1 = Strong
Yuill et al. (2007)	2 = Moderate	2 = Moderate	1 = Strong	3 = Weak	1 = Strong	2 = Moderate

type of schoolyard intervention in relation to social well-being, and previous research has indicated that conflicts are more frequent in schoolyards lacking adequate equipment and age-appropriate options (Doll & Sarver, 2014; Pawlowski et al., 2018). Farmer et al. (2017) found that modifying schoolyards to increase opportunities for risk and challenge led to positive outcomes in children's interactions, despite more bullying. The authors suggest this might be because children in intervention schools became more accustomed to pushing and shoving, indicating greater resilience (Farmer et al., 2017). This unexpected positive effect highlights the potential benefits of designing schoolyards that encourage risk and challenges, which has been recommended for its positive impact on social well-being and behaviours (Brussoni et al., 2015). Unfortunately, such opportunities are increasingly limited by policies prioritising injury prevention and risk reduction over attractive play opportunities (Jerebine et al., 2024).

Previous reviews indicate that greening schoolyards might improve children's social well-being (Bikomeye et al., 2021; Schipperijn et al., 2024). Also, a study by Dymont and Bell (2008) showed that children who play in green schoolyards improved civil and cooperative play compared to those who play on paved schoolyards (Dymont & Bell, 2008). However, in the context of this review, it was not possible to draw definitive conclusions regarding the impact of schoolyard greening on social well-being, as the four studies investigating this relationship yielded conflicting results. Two studies reported positive effects, such as increased prosocial interactions and decreased antisocial behaviour (Bohnert et al., 2022; Raney et al., 2021; Raney et al., 2019), one study found a positive impact on the younger and a negative impact on older children's prosocial orientation (van Dijk-Wesselius et al., 2018) and the last study found a positive improvement in peer play after one year but after two years more bullying were observed by teachers (Farmer et al., 2017).

None of the studies that added markings to schoolyards demonstrated significant effects on social well-being (Crust et al., 2014; Lateef et al., 2024; Mayfield et al., 2017). Although a recent review found that playground markings can boost physical activity during playtimes (Schipperijn et al., 2024), they do not seem to improve social behaviour.

4.2. Emotional well-being

Five studies were identified that looked at emotional well-being, with four out of five studies not detecting overall statistically significant changes. Baseline assessments in the studies by van-Dijk et al. (2018), Lateef et al. (2024), Crust et al. (2014), and Redondo-Tebar et al. (2023) all showed few to no emotional problems among the children, possibly indicating a ceiling effect, which might explain the results. The study by Kelz et al. (2015) did show significant improvements after a combination of greening initiatives and play structures for adolescents attending middle-school in a rural part of Austria. The intervention consisted of age-appropriate structure such as table tennis, volleyball nets, soccer goals, seatings, and greening initiatives such as scrubs and pot plants (Kelz et al., 2015). This suggests that diverse facilities may positively impact emotional well-being for this age group, which is supported by previous research that found positive links between natural environments and emotional well-being (Caryl et al., 2024; Lousen & Andkjær, 2024; McCormick, 2017; Mygind et al., 2019; Roberts et al., 2020).

Some gender differences were observed in the intervention outcomes (Redondo-Tebar et al., 2023). This gender difference warrants attention, as established patterns show that girls are at higher risk of experiencing mental health problems (WHO, 2025). Schoolyard interventions should consider gender-specific needs, ensuring activities appeal to both boys and girls to maximise their potential for promoting well-being (Ly & Vella-Brodrick, 2024).

4.3. Cognitive well-being

The studies indicated a tendency for interventions that combined schoolyard modifications with education or facilitation to have a more positive impact on cognitive well-being (Khan et al., 2020; Sánchez-López et al., 2019) than modifications alone (Kelz et al., 2015; van Dijk-Wesselius et al., 2018). Supporting this, other studies found that children participating in outdoor learning programs preferred outdoor education over traditional indoor learning (Athman & Monroe, 2004; Singal & Swann, 2011). However, van Dijk-Wesselius et al. (2018) found a significant improvement in attentional functioning over time with just greening schoolyards. Other studies have also found playing in green schoolyards to positively impact pupils' academic performance (Amholt, 2022; Amicone et al., 2018; Lopez et al., 2008). This finding is consistent with Attention Restoration Theory (Kaplan & Kaplan, 1989) suggesting that natural environments restore directed attention, and Stress Reduction Theory (Ulrich, 1991), which indicates that exposure to nature reduces stress. Furthermore, a recent study found the greatest benefits among pupils from urbanised areas, where the schoolyard served as a valuable source of green space, compensating for limited exposure to nature (Aerts et al., 2025). Though, studies assessing the effect of schoolyard modifications on cognitive well-being is limited, and further research is necessary to determine whether schoolyard interventions can enhance cognitive well-being, with or without teacher facilitation.

4.4. Limitations

4.4.1. Limitations of the reviewed evidence

The included studies were characterised by variation in study designs, and in particular substantial variation in measurement methods and outcomes, which made an overall synthesis of results impractical. Several studies used validated instruments only, but others used personally developed measurement tools with unclear validity. Since well-being is a complex issue with multiple components that can be measured in many ways, harmonisation is difficult. Another limitation is the considerable variation between schools, including differences in type, cohorts, and available resources, which may have influenced both the implementation and the reported outcomes.

4.4.2. Limitations of the review process

Only scientific databases were searched, as the review was initially planned as a systematic review. Grey literature was not included based on librarian guidance and the assumption that relevant peer-reviewed studies are well indexed in the selected databases. For the screening process ASReview was utilised. In the context of the expanding number of publications, researchers often develop narrow searches to manage workload, potentially missing relevant studies (Miwa et al., 2014; Schoot et al., 2021). The time-saving advantages of ASReview enabled us to develop a search strategy focused exclusively on research relevance, without workload constraints. While accepting a risk of missing up to 5 % of relevant records, research suggests that traditional manual screening carries similar risks due to reviewer fatigue and human error (Campos et al., 2024). Although alignment between the two independent reviewer groups was not optimal, double-screening and merging the datasets before full-text screening enhanced validity compared to screening by only one group.

4.5. Implications for future research

It would be highly beneficial if future studies would use standardised, validated measures for the well-being construct they are investigating, and report their effect estimates enabling meta-analysis. One way to select a validated measurement method is to consult Table 2 of this paper. We also recommend that researchers carefully select instruments that are sensitive to the ceiling effect, particularly

when using self-rated measures of emotional well-being. Furthermore, sample sizes should be sufficiently large to be able to detect differences between groups.

Most studies included children aged five to 13, which reflects the interest in schoolyards for younger children (Amholt, 2022). Future research should also examine the benefits of different schoolyard features in supporting the well-being of adolescents.

4.6. Implications for school health policy, practice, and equity

There is unlikely to be a single ‘silver bullet’ for improving well-being in children and adolescents through schoolyard interventions alone. Instead, to achieve lasting benefits, such efforts should be embedded within broader systems-based or ecological approaches. Global physical activity policy frameworks, such as the World Health Organisation’s Global Action Plan on Physical Activity (GAPPA) and the International Society for Physical Activity and Health’s Eight Investments, which emphasise multi-level action, can offer valuable guidance for the development of future interventions. It is also important to consider the cost of schoolyard modifications to avoid inadvertently widening inequalities, particularly if less advantaged schools are unable to afford such improvements.

Nevertheless, based on the limited evidence, some tentative conclusions can be drawn from this review, supported by previous research:

- A greater variety of play structures in the schoolyard environment may positively influence both social and emotional well-being.
- Spending time in green areas in the schoolyard can improve emotional well-being, however, this review only demonstrated this effect in adolescents aged 13–15 years.
- Interventions combining schoolyard modifications with educational components may result in larger cognitive well-being benefits.

5. Conclusion

Research on how schoolyard interventions impact social, emotional, and cognitive well-being has increased over the past two decades, yet it remains limited with only 13 studies (14 articles) identified. In this review, we examined studies that quantitatively evaluated changes in schoolyards, including play structures, greening, or markings, and their effects on children’s and adolescents’ well-being. The included studies varied widely in terms of interventions, study designs, reporting of effects, and quality. Furthermore, this review shows that well-being is a complex field, with the 13 studies reporting on 31 different outcomes measured with 18 different measurement tools.

Most studies yielded non-significant or conflicting findings, however, findings tentatively indicated that a greater variety of play structures, spending time in green areas, and adding play structures combined with educational components tended to enhance social, emotional, and cognitive well-being. The evidence base remains limited and fragmented, highlighting the need for further research based on robust study designs, methodology, and sample sizes sufficient to detect differences.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Copilot in order to support language refinement, primarily by drawing inspiration from its suggestions for phrasings and formulations. After using the tool, the authors reviewed and edited the content and take full responsibility for the content of the published article.

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Declaration of competing interest

All authors declare that they have no conflicts of interest.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.mhp.2025.200460.

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